WORKPULSE: EMPLOYEE ATTENDANCE AND WORK ANALYTICS INFORMATION SYSTEM FOR MVL

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Chapter 1

INTRODUCTION

Background of the Study

Employee attendance and work information management is a vital component of organizational operations, as it directly affects productivity, accountability, and decisionmaking. Globally, organizations are increasingly adopting digital and analytics-driven human resource systems to replace manual and outdated attendance methods that often result in errors, inefficiencies, and limited access to actionable insights. Recent literature highlights that the integration of biometric-based system is revolutionizing human resource management by enhancing operational efficiency, minimizing labor-related inefficiencies, and eliminating the administrative burdens of traditional tracking methods (ljiga et al., 2025). Modern workforce analytics systems enable organizations to record attendance accurately, monitor work patterns, and generate data-driven reports that support effective human resource management. Despite these global advancements, studies reveal that many organizations still rely on disconnected or partly manual processes to manage employee information, attendance, leave, and payroll. These practices often result in inconsistent records, delayed approvals, limited employee transparency, inefficient reporting, and weak accountability. Basic systems may also lack appropriate authentication, authorization, audit logging, and protection for sensitive employee information, which can compromise employee trust and organizational credibility. Furthermore, traditional payroll and security systems such as manual logs or PIN codes contribute heavily to human error and unauthorized entry, explicitly highlighting the critical need for biometric verification to eliminate proxy attendance practices like "buddy punching" (International Journal of Science and Research Archive, 2026). In the Philippine context, organizations are rapidly transitioning toward digital transformation in response to the growing demand for efficient, transparent, and technology-driven workplace systems. The Philippine HR technology market is experiencing significant growth as employers increasingly rely on cloud-based platforms and digital portals for faster, more efficient workforce processing (IMARC Group, 2025). This push for digitalization is evident even at the national level, with recent milestone initiatives like the Philippine Civil Service Modernization Project launched in late 2025, which seeks to harness digital tools to automate human resource and payroll systems for

better efficiency and transparency (Civil Service Commission, 2025). However, many private and local organizations continue to face challenges such as fragmented attendance records, inefficient monitoring of work hours, and limited analytical capability. A centralized and secure workforce information management system is therefore needed to improve record management, administrative workflows, employee access, and data-supported decision-making. In response to these challenges, this study proposes WorkPulse, a secure web-based workforce information management system with integrated fingerprint-based attendance verification. The project aims to centralize employee, attendance, leave, and payroll records while providing strict role-based access organized into administrator and employee access groups. By utilizing a fingerprint scanner to verify employee identity during time-in and timeout, the system aims to associate biometric attendance records with the verified employee, improving identity assurance and operational integrity. Data is securely managed using a cloud-hosted MongoDB Atlas platform to support centralized, authorized data storage and access. This study aligns with Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth by promoting fair labor practices, and SDG 9: Industry, Innovation, and Infrastructure by integrating digital analytics and secure hardware solutions into organizational process.

Statement of the Problem

Many organizations manage employee information, attendance, leave, and payroll through disconnected or partly manual processes. These practices often lead to inaccurate records, lack of transparency, inefficient workforce planning, and weak security controls that may expose sensitive employee data. In view of the foregoing, this study seeks to answer the following questions: 1. What challenges are encountered in managing employee information, attendance, leave, and payroll through current manual or fragmented processes? 2. What functional and security requirements should WorkPulse provide for administrators and employees? 3. How can the system centralize workforce records and support attendance monitoring, leave processing, payroll workflows, and employee self-service? 4. How acceptable is the developed system in terms of functional suitability, usability, performance efficiency, reliability, and security? 5. How accurately and reliably does the integrated fingerprint verification authenticate employee attendance?

Objectives of the Study

The primary objective of this study is to design, develop, and evaluate WorkPulse, a secure web-based workforce information management system that centralizes employee records, attendance, leave, cash advance requests, and payroll workflows using MongoDB Atlas; provides administrator and employee access groups; supports fingerprint and authorized manual attendance processes; and offers workforce reporting and implemented analytical features.

The specific objectives of this study are:

1. To identify problems and requirements in existing employee information, attendance, leave, cash advance, and payroll processes.

2. To develop a centralized web-based system integrated with MongoDB Atlas for employee records, attendance monitoring, leave management, cash advance processing, and payroll workflows.

3. To implement administrator and employee access groups with authenticated access, record ownership controls, and support for Regular, Extra, and Manager / Supervisor employee classifications.

4. To provide an employee portal for viewing authorized personal, attendance, payroll, and leave information; submitting leave and cash advance requests; and receiving employee notifications.

5. To implement appropriate security controls, including authentication, authorization, session management, request protection, input validation, audit logging, and controlled administrator attendance corrections with required reasons, correction history, and protection of attendance already included in settled payroll.

6. To generate descriptive dashboards and reports from stored workforce records and provide only those analytical forecasting functions that are implemented and evaluated; to support limited offline saving and later synchronization for designated activities.

7. To evaluate the system using defined software-quality criteria and test procedures, including the operation of employee notifications, cash advance workflows, manual attendance corrections, and offline synchronization where implemented.

8. To integrate and evaluate fingerprint-based time-in and time-out verification using measured biometric performance and usability results.

Scope and Delimitation

This study covers the design and development of WorkPulse as a responsive, web-based workforce information management system using a cloud-hosted MongoDB Atlas database. The system has two access groups: administrator and employee. Employee classifications consist of Regular, Extra, and Manager / Supervisor. These classifications describe employees and do not establish additional system access groups or a separate management portal.

The attendance component includes time-in and time-out through a configured fingerprint attendance kiosk, as well as authorized administrator entry of manual attendance and corrections. Administrators must provide a reason for each correction, and the system retains a correction history for accountability. Attendance records already included in settled payroll are protected from direct editing; any necessary adjustment must follow an authorized, traceable procedure rather than silently changing settled records.

The employee portal supports viewing authorized personal information, attendance, leave, payroll and payslip summaries, submitting leave requests and cash advance requests, and receiving employee notifications about relevant actions or status changes. Cash advance workflows cover request submission, administrative review, disbursement confirmation, outstanding balance tracking, repayment deductions, carried balances between payroll periods, and related payslip summaries. Payroll processing is subject to authorized review and settlement controls.

The system includes authentication and email OTP where configured, role-based access control, employee information management, system settings, audit logging, and descriptive workforce dashboards. Descriptive reporting summarizes recorded events and totals; analytical forecasting is a distinct function. The forecasting scope is limited to the implemented and evaluated functions: estimates of expected employee attendance over the next seven days, estimates of attendance-risk events over the next 30 days, and identification of unusual arrival patterns. Forecasting outputs are not equivalent to measured employee productivity, and any statistical fallback must be distinguished from a trained machine-learning result.

The system provides limited offline saving for designated manual attendance entries or corrections, leave requests, cash advance requests, and viewing information previously cached on the device. Eligible offline actions are held in an offline queue and synchronized when connectivity returns, subject to server-side validation, authorization, and conflict handling.

Fingerprint-based attendance requires the supported fingerprint scanner connected to the configured local attendance workstation, together with its reader and matching software. Fingerprint capture and verification cannot be performed solely through a remote cloud browser session. The system does not include facial recognition or webcam-based attendance verification.

Cloud access to WorkPulse is separate from the local fingerprint workstation: web access to the deployed application and synchronized MongoDB Atlas records depends on the configured server, network availability, authentication services, and browser compatibility. Limited offline support does not make the entire system fully operational without internet connectivity. Fingerprint scanning, payments or disbursement execution, approval decisions, and fresh login require connectivity. Offline entries remain pending until successful synchronization and server-side checks.

WorkPulse supports workforce reporting and only the forecasting functions actually implemented and evaluated; it does not provide a complete measurement of individual employee productivity. Cash advance tracking and payroll summaries do not themselves constitute integration with a banking or payment gateway. The study does not claim biometric accuracy, forecasting performance, or offline reliability beyond the results demonstrated in the corresponding testing chapters.

Significance of the Study

This study aims to contribute to improved workforce management and organizational efficiency. The system may benefit the following: Administrators and Management. Administrators can efficiently track verifiable employee attendance, manage leave approvals, process payroll workflows, and make informed decisions using centralized data dashboards hosted on a scalable cloud database. The system helps centralize workforce records, reduce repetitive administrative work, and provides a highly reliable method of confirming physical attendance through biometric integration. Employees. Employees benefit from accurate

attendance records, fair evaluations, and an intuitive self-service portal where they can view their own authorized records, monitor their payroll, and submit leave requests, improving transparency. IT Practitioners and Developers. IT Practitioners and Developers: The project demonstrates biometric hardware integration, cloud database integration using MongoDB Atlas, authentication, authorization, audit logging, multi-factor authentication (MFA), and other security controls relevant to modern web-based workforce systems. Future Researchers. The system provides a foundation for future studies involving biometric attendance integration, cloud-based workforce analytics, privacy-aware management systems, and the evaluation of Al-assisted administrative tools.

Definition of Terms

The following terms are operationally defined as used in this study:

Attendance Correction. An authorized change to an attendance record, including a manual adjustment, for which an administrator records a reason and the system retains a traceable correction history. Records included in settled payroll cannot be directly overwritten.

Attendance Monitoring. The viewing, tracking, and management of employee attendance records, including time-in, time-out, authorized manual entries, and corrections.

Audit Logging (Security Audit Logging). The recording of important user, administrative, and security events, including attendance corrections, to support accountability and review.

Authentication. The process of verifying a user's identity before permitting access to WorkPulse.

Authorization. The determination of which features and records an authenticated user may access based on their access group and record ownership.

Biometric Verification. The use of a fingerprint captured by the configured scanner to verify an employee's identity for supported attendance transactions.

Carried Balance. The remaining unpaid amount of a cash advance that is transferred to a subsequent payroll period after applicable repayments are recorded.

Cash Advance. An employee request for an advance amount, subject to administrative review, confirmed disbursement, balance tracking, and repayment recording.

Cloud Database (Cloud-Based Data Storage). The remote database infrastructure used to store authorized WorkPulse workforce records; the project uses MongoDB Atlas.

Dashboard. An interface presenting summarized workforce information through indicators, tables, charts, and reports.

Data Validation. Checking entered or synchronized information against required formats and business rules to reduce invalid records.

Descriptive Analytics. The summarization of historical or current workforce records using counts, tables, charts, and reports without forecasting future outcomes.

Employee. An authorized user in the employee access group who may be classified as Regular, Extra, or Manager / Supervisor and may access permitted self-service features.

Employee Information Management. The centralized organization and maintenance of employee profiles and related workforce records by authorized users.

Employee Notification. A system-generated notice informing an employee about relevant record updates, request decisions, or other actions available in WorkPulse.

Employee Portal (Employee Self-Service Portal). The employee-facing interface for viewing authorized personal, attendance, leave, payroll, and payslip information, submitting requests, and receiving notifications.

Encryption. A security technique that transforms information to help protect it from unauthorized access.

Fingerprint Scanner. The supported local biometric hardware device used to capture fingerprint data for verification.

Fingerprint Verification. The process of comparing a captured fingerprint against authorized reference data to verify an employee during biometric attendance.

Fingerprint-Based Attendance. An attendance transaction that uses the configured fingerprint reader and matching software to verify identity before recording time-in or time-out.

Leave Management. The submission, review, approval or rejection, and recording of employee leave requests.

Manager / Supervisor. An employee classification within WorkPulse, alongside Regular and Extra; it does not automatically create a separate system access group.

MongoDB Atlas. The cloud-hosted database platform used to manage WorkPulse workforce records.

Multi-Factor Authentication (MFA). An authentication approach requiring more than one verification factor; WorkPulse incorporates email OTP as an additional access measure where configured.

Offline Queue. Temporary local storage of eligible actions created without connectivity and awaiting validation and synchronization with the server.

One-Time Password (OTP). A temporary code sent to an authorized email address for an applicable authentication step.

Payroll Management (Payroll Workflow). The authorized recording, review, calculation-related processing, and settlement of payroll information, including supported cash advance repayments and carried balances.

Payroll Summary / Payslip View. The employee-accessible display of authorized pay information, including relevant cash advance deductions and remaining balances.

Performance Efficiency. The ability of the system to respond to user actions and process supported tasks without unnecessary delays.

Predictive Insights (Analytical Forecasting). Outputs of implemented and evaluated workforce analytical functions that estimate future attendance or attendance-risk events or flag unusual arrival patterns. They are separate from descriptive reports and are not a direct measurement of productivity.

Role-Based Access Control (RBAC). A security control that limits access according to the administrator or employee access group and applicable record permissions; Regular, Extra, and Manager / Supervisor are employee classifications.

Secure Authentication. Verification of user identity through configured controls such as credentials and email OTP.

Session Management. The creation, validation, maintenance, and termination of authenticated user sessions.

Synchronization. The transfer and reconciliation of eligible offline queued actions with the server once connectivity returns, subject to authorization and validation.

System Enhancement. The improvement or addition of system functions, controls, or other capabilities.

System Evaluation. The assessment of WorkPulse against defined usability, functional, security, performance, and reliability criteria.

System Quality. The quality of WorkPulse as determined through the stated evaluation criteria and observed results.

System Usability Scale (SUS). A ten-item, five-point questionnaire used to assess perceived system usability.

Time-In and Time-Out. The recording of the beginning and ending of an employee's work period through fingerprint verification or authorized administrator manual attendance, as applicable.

Web-Based Workforce Information Management System. A browser-accessible system for managing employee, attendance, leave, cash advance, payroll, reporting, and security functions.

Weighted Mean. The computed average rating used to summarize responses to a system evaluation criterion.

WorkPulse. The web-based employee attendance and work analytics information system developed for MVL, including the workforce modules and controls defined in this study.

Workforce Analytics Dashboard. A presentation of descriptive workforce indicators and reports, distinct from separately identified forecasting functions.

Workforce Information Management. The centralized organization and processing of employee-related records and workforce operations.

Workforce Planning. The use of workforce information and evaluated analytics to inform organizational staffing and operational decisions.